

Day One Setup Checklist

Name

GitHub username

Operating system

Two things finish this: SETUP CODE: UU-92-5650 on your screen, and your commit in your own fork.

Everyone gets the same setup code. Show it, and your fork on github.com, to the instructor or an IA before you leave. Nothing is graded today, but Thursday's lab assumes all of it works.

1 Make a folder for this class

Somewhere you will find again: Documents, not Downloads. Make a course folder with an empty data folder inside it. By the end of step 5:

```
DS5030/
├── data/           ← datasets you download later in the semester that are too big to push/pull from github
└── UU_F26_templates/ ← step 5 puts your fork here
```

☐ My class folder exists, and it has an empty data folder inside it.

2 Open the folder in VS Code, and add two extensions

File > Open Folder, and choose the folder you just made. Then open Extensions (Cmd/Ctrl + Shift + X) and install both (published by Microsoft):

☐ **Python** installed ☐ **Jupyter** installed

☐ The VS Code sidebar shows my class folder, with data inside it.

3 Create the conda environment

An *environment* is a private Python, so this class cannot break your other projects. No conda yet? Install **Miniconda** from docs.conda.io/en/latest/miniconda.html, then **close and reopen your terminal**. Run these two lines, one at a time:

```
conda create -n uu -c conda-forge python=3.12 numpy "pandas>=2.2,<3" matplotlib \
    seaborn scipy statsmodels networkx ipykernel jupyterlab
conda activate uu
```

☐ It finished without an error (a few minutes), and my prompt now starts with (uu).

4 Point VS Code at that environment

VS Code will not find it on its own, and this is the step people skip. Press Cmd/Ctrl + Shift + P, type **Python: Select Interpreter**, and pick the one labelled uu.

☐ The status bar at the bottom of the VS Code window now shows uu.

5 Fork the repository, then clone your fork

You work in your own copy, not mine. A *fork* is your own GitHub copy of my repository: you push your work to yours, and pull my updates from mine. One fresh repository per lab.

Fork it on the web first. At github.com/smgroves/UU_F26_templates, click **Fork** (top right) and keep the name. You now own github.com/YOUR-USERNAME/UU_F26_templates.

Option A · GitHub Desktop **RECOMMENDED**

Install from desktop.github.com and sign in, which also settles your push permissions. Then **File > Clone repository > GitHub.com**, pick the UU_F26_templates under **your own** username, and set the local path to your class folder. **If it asks how you plan to use the fork, choose the option about your own purposes**, not contributing to the parent: that keeps your pushes going to your fork, not mine.

Option B · Command line

In VS Code's terminal (**Terminal > New Terminal**), from your class folder, with **your own username**:

```
git clone https://github.com/YOUR-USERNAME/UU_F26_templates.git
```

Then everyone, in VS Code's terminal, add my copy as upstream and check both remotes:

```
cd UU_F26_templates
git remote add upstream https://github.com/smgroves/UU_F26_templates.git
git remote -v
```

☐ origin shows **my own username** (where my work goes) and upstream shows smgroves (where updates come from).

☐ UU_F26_templates is in the VS Code sidebar next to data, and contains python_test.ipynb.

6 Run the notebook, and check every line

Open `UU_F26_templates/python_test.ipynb` from the sidebar. **Look at the top right: the kernel must say uu.** If it says base, Python 3, or Select Kernel, click it and pick uu. Then **Run All**, and tick a box for each package that printed a version number:

- ☐ numpy ☐ pandas ☐ matplotlib ☐ seaborn ☐ scipy ☐ statsmodels
- ☐ The interpreter line contains `/uu/`. If not, the kernel is wrong: fix it before going on.
- ☐ **A histogram appeared.** Not a warning, not a blank space: an actual picture.
- ☐ The last line reads exactly `SETUP CODE: UU-92-5650`.

7 Push the finished notebook to your fork

Save the notebook first (Cmd/Ctrl + S), so the run you just did is on disk.

Option A · GitHub Desktop **RECOMMENDED**

It already lists `python_test.ipynb` as changed. Type a summary such as `setup complete`, click **Commit to main**, then **Push origin**.

Option B · Command line

```
git add python_test.ipynb
git commit -m "setup complete"
git push
```

- ☐ I opened `github.com/YOUR-USERNAME/UU_F26_templates` in a browser and **my commit is there**.
- ☐ I showed that page, and the setup code, to the instructor or an IA.

Pulling my updates later. Easiest route: click **Sync fork** on your fork's github.com page, then pull. From the terminal: `git pull upstream main`. Do this before every class. **Next week's lab is group work**, and we will add collaboration then.

If it breaks

"conda: command not found": you did not reopen the terminal after installing Miniconda. On Windows, use the **Anaconda Prompt** rather than the default terminal.

The kernel picker does not offer uu: quit VS Code entirely and reopen. Still missing? You left ipykernel out of step 3: run `conda install -n uu ipykernel`.

No /uu/ in the interpreter line: the notebook is running a different Python than your packages went into. **The most common problem in this class**, and fixing it now beats fixing it Thursday. Redo step 4, then re-pick the kernel.

origin says smgroves: you cloned mine instead of your fork. Repoint it with `git remote set-url origin https://github.com/YOUR-USERNAME/UU_F26_templates.git`.

"git: command not found": install GitHub Desktop, which includes git. On a new Mac you can instead run `xcode-select --install` and wait for it to finish.

The push is rejected, or asks for a password: GitHub stopped accepting passwords in 2021. Use GitHub Desktop, which handles this for you, or set up a personal access token or SSH key. Come find us rather than fighting it.

"No such file or directory: cville_cars.csv": the notebook has to run from inside `UU_F26_templates`. Open it from the sidebar rather than moving the file.

Fallback: Google Colab

If your machine will not install anything, you can still prove the code works in the browser. Go to `colab.research.google.com`, then **File > Open notebook > GitHub**, and paste your own fork:

```
YOUR-USERNAME/UU_F26_templates
```

Open `python_test.ipynb`, add **one new cell at the very top**, and run it first so the data file is there:

```
!git clone https://github.com/YOUR-
  USERNAME/UU_F26_templates.git
%cd UU_F26_templates
```

Colab has all six packages, so you should reach the same `UU-92-5650`. Tick the step 6 boxes.

Step 7 is the catch: pushing from Colab is awkward. Use **File > Save a copy in GitHub**, pointing at your own fork, and **tell us you are on Colab today** so we can make sure the labs work for you before they count.

Before Thursday (Aug 27): watch the data wrangling video, and finish this checklist if you got stuck — reach out with questions over email or Piazza. Thursday works in the same folder, in the same environment, so an unfinished setup is the one thing that will put you behind.